

# Near-strict 1-spike-per-cycle ceiling: 98.9% of (cell, cycle) pairs

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## Abstract

exp041's slope  $p \approx 0.20$  was interpreted as “each E cell joins a cycle with  $\approx 20\%$  probability”. That reading assumes the per-cycle spike count is bounded by 1 (an E cell either participates in this cycle or not). This notebook measures that directly: walking through every gamma cycle on the test set and counting how many spikes each E cell actually emits, on all 18 trained checkpoints in exp041's  $\tau_{\text{GABA}}$  sweep.

## Methods

For each of exp041's 18 trained cells ( $6 \tau_{\text{GABA}} \times 3$  seeds):

1. Run inference on the MNIST test set; capture per-trial  $(T, B, N_E)$  and  $(T, B, N_I)$  spike tensors.
2. Detect I-burst times per trial: smooth the population I rate with a 1-ms Gaussian, run scipy peak detection with min-distance set to half the cell's own  $1/f_\gamma$ .
3. Cycle boundaries are the midpoints between consecutive I-burst peaks (first cycle starts at  $t = 0$ , last ends at trial end).
4. For each (cell, cycle, trial), count the number of E spikes within the cycle window.
5. Bucket counts globally into  $\{0, 1, 2, \geq 3\}$  and aggregate by  $\tau_{\text{GABA}}$ .

The cycle anchor is the I-burst: this is the right anchor because the cycle is operationally defined as “the time between one inhibitory blanket and the next”, not as the time between E bursts (which can be silent on a given cycle).

## Results

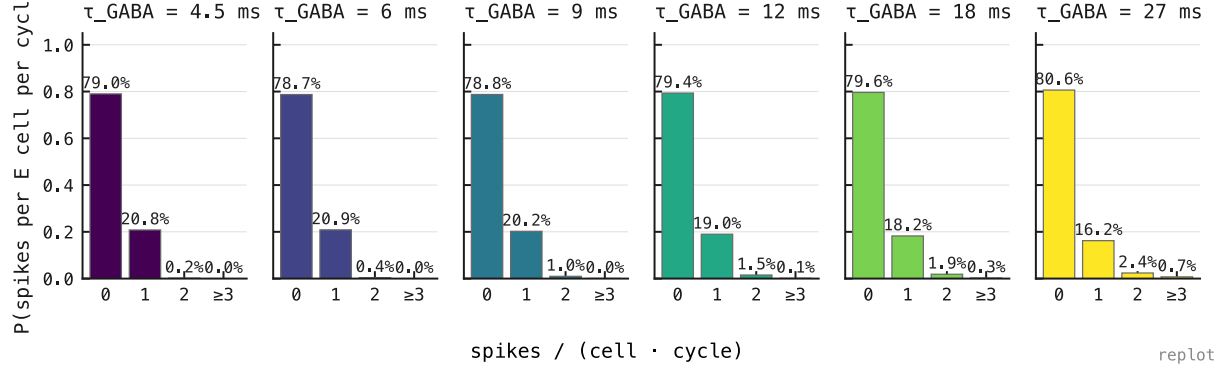


Figure 1: Distribution of E spike count per gamma cycle per cell, by  $\tau_{\text{GABA}}$ , three seeds aggregated. Across **179 million (cell, cycle) pairs**, the architecture is overwhelmingly bimodal: each cell either emits zero spikes in a given cycle ( $\approx 79\%$  of the time) or exactly one ( $\approx 20\%$  of the time). Two-or-more events occur in  $\approx 1.1\%$  of cycles; three-or-more in  $\approx 0.14\%$ . Pooled over the sweep, 98.9% of pairs carry at most one spike.

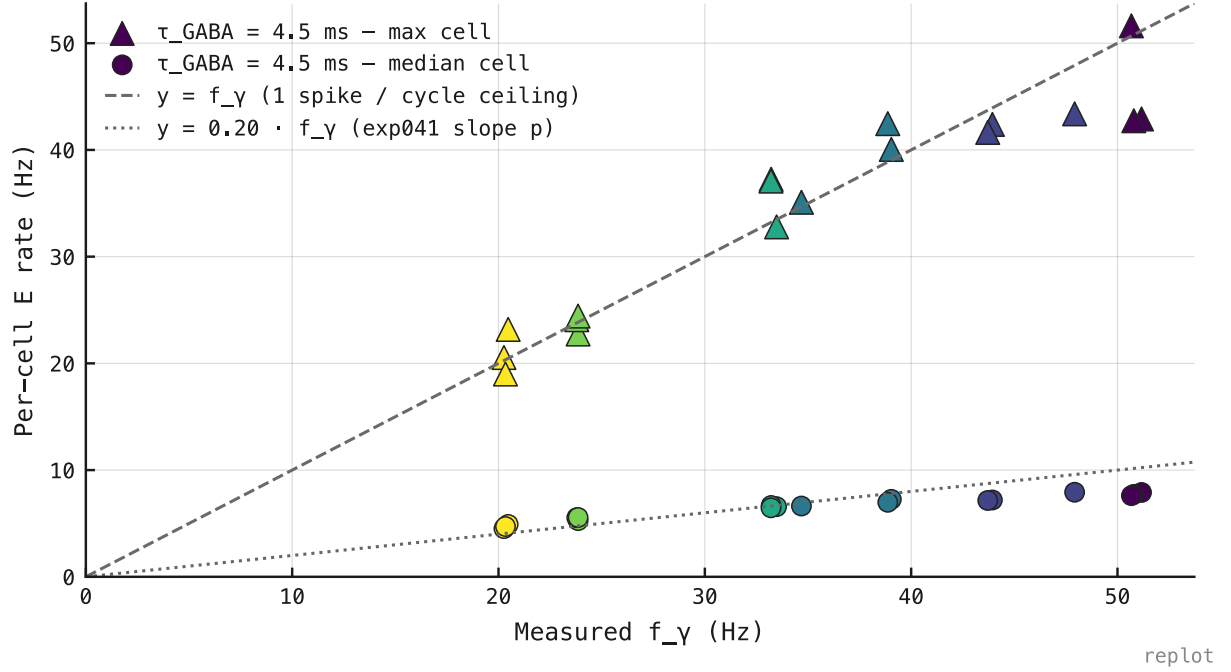


Figure 2: Per-cell E rate versus measured gamma frequency  $f_{\gamma}$  across the  $\tau_{\text{GABA}}$  sweep. The busiest cell in each network tracks the one-spike-per-cycle ceiling  $r = f_{\gamma}$  (max-cell fit  $r = 0.97f_{\gamma}$ ,  $R^2 = 0.88$ ), while the median cell sits on exp041's shallower  $r \approx 0.20f_{\gamma}$  participation slope. The ceiling is near-strict: even the most active cell rarely exceeds one spike per cycle.